



Computer Systems Security

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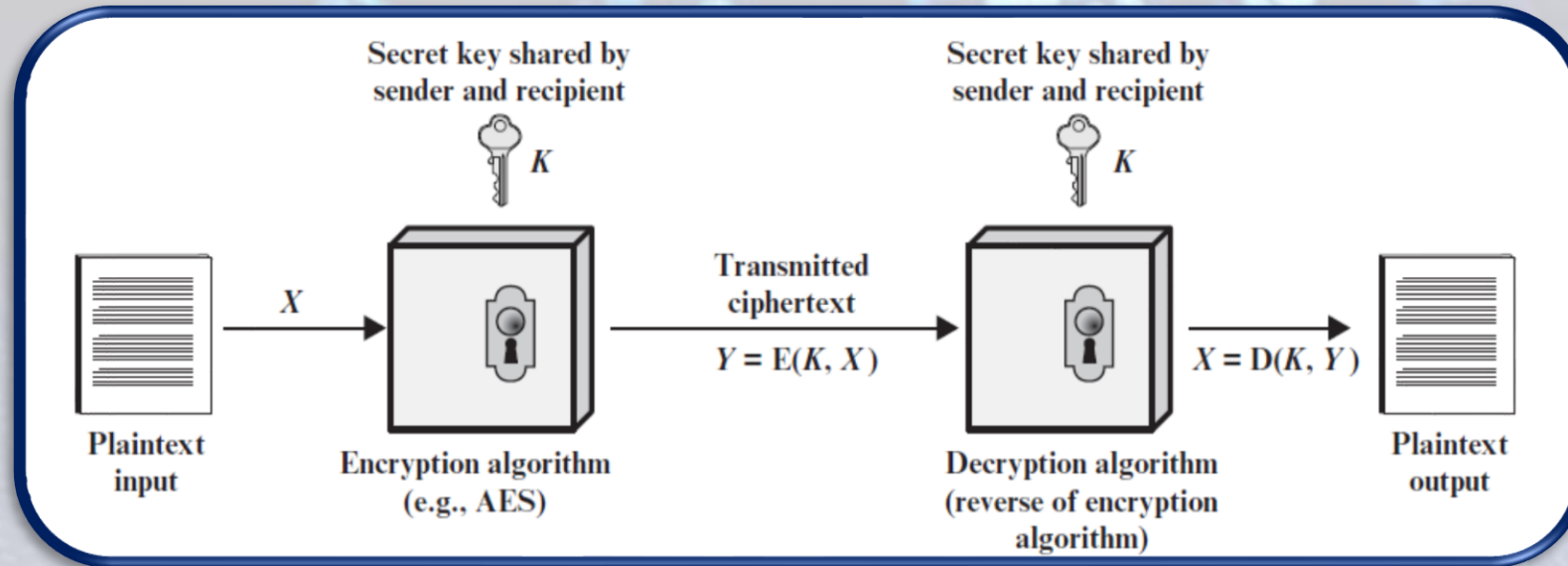
Symmetric Cipher Model/Symmetric Encryption

- Also referred to as conventional encryption or **single-key** encryption.
- Prior to the development of **public-key** encryption.
- The **most widely used** of the two types of encryption.
- A symmetric encryption scheme has five ingredients:
 1. **Plaintext:** is the original message or data that is fed into the algorithm as input.
 2. **Encryption algorithm:** performs various substitutions and transformations on the plaintext.
 3. **Secret key:** is also input to the encryption algorithm. The key is a value **independent** of the plaintext and of the algorithm. The algorithm will produce a *different* output depending on the specific key being used at the time. The exact substitutions and transformations performed by the algorithm depend on the key.
 4. **Ciphertext:** is the scrambled message produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different ciphertexts.
 5. **Decryption algorithm:** is essentially the encryption algorithm run in reverse. It takes the ciphertext and the secret key and produces the original plaintext.

Some Terminology

- **Enciphering or encryption:** is the process of converting from plaintext to ciphertext.
- **Deciphering or decryption:** is the process of restoring the plaintext from the ciphertext.
- The many schemes used for encryption constitute the area of study known as **cryptography**.
- Techniques used for deciphering a message without any knowledge of the enciphering details fall into the area of **cryptanalysis** known as “breaking the code”.
- The areas of *cryptography* and *cryptanalysis* together are called **cryptology**.

Symmetric Encryption



- There are **two** requirements for secure use of conventional encryption:
 1. **Strong encryption algorithm:** The opponent should be unable to *decrypt ciphertext or discover the key* even if he or she is in possession of a number of ciphertexts together with the plaintext that produced each ciphertext.
 2. **Secret key:** Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key *secure*. If someone can discover the key and knows the algorithm, all communication using this key is readable.

Cryptography

- Cryptographic systems are characterized along **three** independent dimensions:

Type of operations:

- substitution (each element in the plaintext is mapped into another element).
- transposition (elements in the plaintext are rearranged).
- product systems (multiple stages of substitutions and transpositions).

The number of keys used:

- sender and receiver use the same key: symmetric, single-key, secret-key, or conventional encryption.
- sender and receiver use different keys: asymmetric, two-key, or public-key encryption.

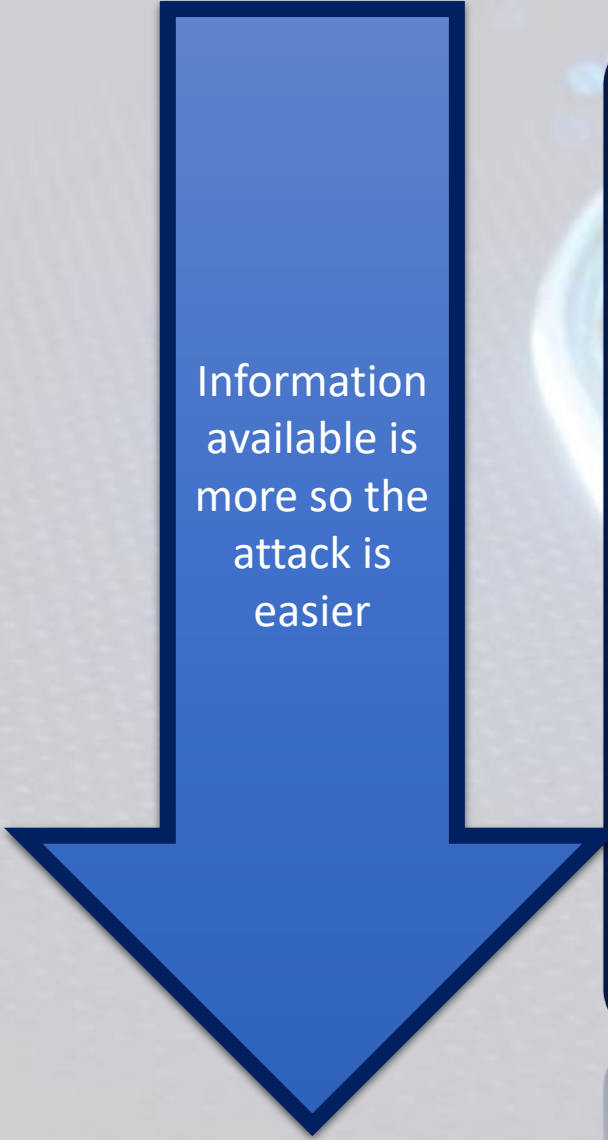
The way in which the plaintext is processed:

- block cipher: processes the input one block of elements at a time, producing an output block for each input block.
- stream cipher: processes the input elements continuously, producing output one element at a time, as it goes along.

Cryptanalysis and Brute-Force Attack

- Typically, the objective of attacking an encryption system is to **recover the key** in use rather than simply to recover the plaintext of a single ciphertext.
- There are **two** general approaches to attacking a conventional encryption scheme:
 - **Cryptanalysis**: This type of attack exploits the characteristics of the ***algorithm*** to attempt to deduce a specific plaintext or to deduce the key being used.
 - **Brute-force attack**: The attacker tries ***every possible key*** on a piece of ciphertext until an **intelligible** translation into plaintext is obtained. On average, half of all possible keys must be tried to achieve success.
 - To supplement the brute-force approach, some degree of knowledge about the expected plaintext is needed, and some means of automatically distinguishing plaintext from garble is also needed, e.g.: problem is more difficult if the data is *compressed* or *numerical*.

Types of Cryptanalytic Attacks



Information available is more so the attack is easier

Type of Attack	Known to Cryptanalyst
Ciphertext Only	■ Encryption algorithm ■ Ciphertext
Known Plaintext	■ Encryption algorithm ■ Ciphertext ■ One or more plaintext–ciphertext pairs formed with the secret key
Chosen Plaintext	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	■ Encryption algorithm ■ Ciphertext ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Some Definitions

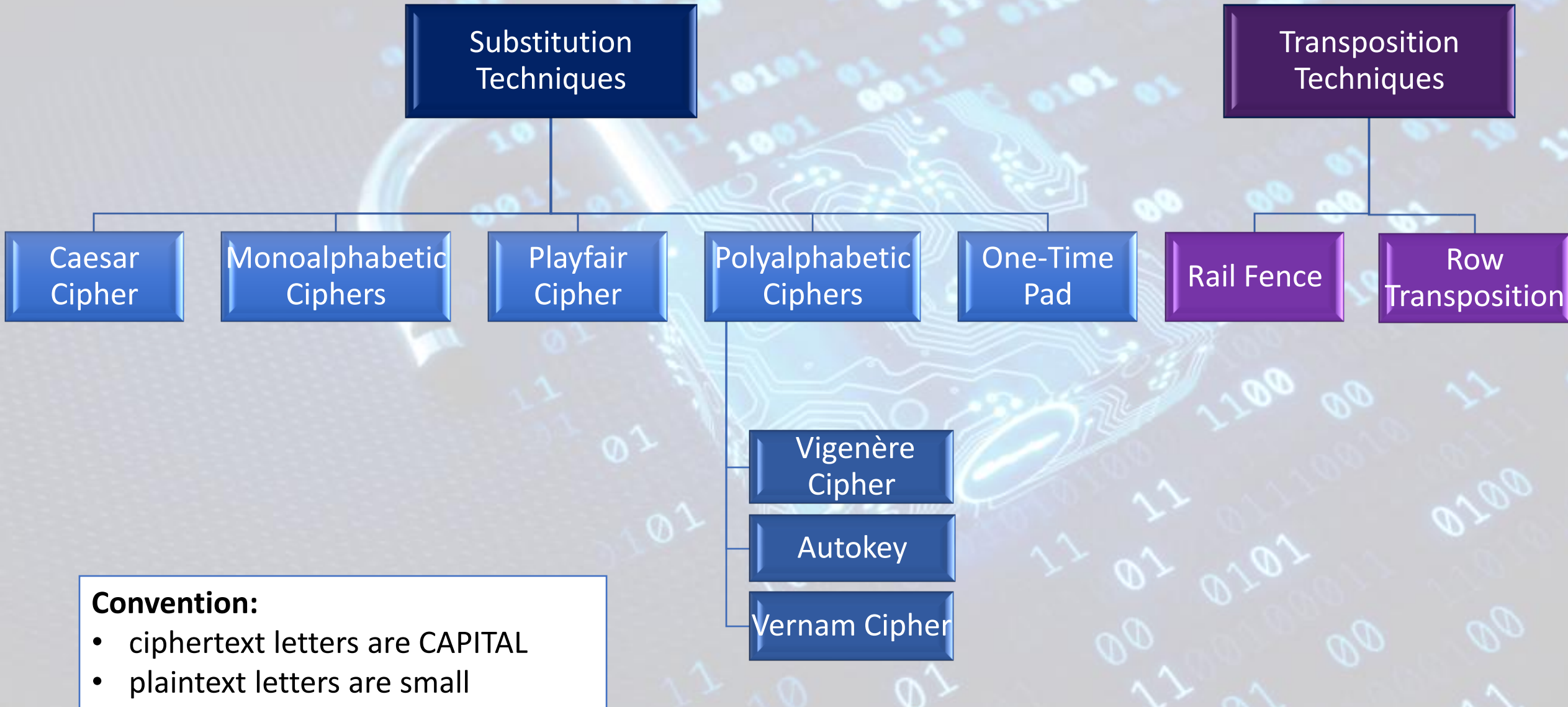
- Unconditionally secure:

- No matter how much computer *power* or *time* is available, the cipher cannot be broken since the ciphertext provides insufficient information to uniquely determine the corresponding plaintext.
- There is no encryption algorithm that is unconditionally secure (except one-time pad).

- Computationally secure:

- Therefore, the goal is a computationally secure encryption algorithm that meets one or both of the following criteria:
 - ❑ The **cost of breaking** the cipher exceeds the value of the encrypted information.
 - ❑ The **time required** to break the cipher exceeds the useful lifetime of the information.

Classical Symmetric Encryption Techniques



Caesar Cipher

- Involves replacing each letter of the alphabet with the letter standing three places further down the alphabet.
- Note that the alphabet is wrapped around, so that the letter following Z is A.

plain: meet me after the toga party
cipher: PHHW PH DIWHU WKH WRJD SDUWB

- Listing all possibilities:

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z
cipher: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

- The algorithm can be expressed as follows: $C = E(3, p) = (p + 3) \bmod 26$
- Where the alphabet is numbered as follows:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

- The general Caesar algorithm:
- The decryption algorithm:

$$C = E(k, p) = (p + k) \bmod 26$$

$$p = D(k, C) = (C - k) \bmod 26$$

where k is in the
range 1 to 25

Cryptanalysis of Caesar Cipher

- If it is known that a given ciphertext is a Caesar cipher, then a brute-force cryptanalysis is easily performed: simply **try all the 25 possible keys**.

- Example:

KEY	PHHW PH DIWHU WKH WRJD SDUWB		
1	oggv og chvgt vjg vqic retva	14	btti bt puitg iwt idvp epgin
2	nffu nf bqufs uif uphb qbsuz	15	assh as othsf hvs hcuo dofhm
3	meet me after the toga party	16	zrrg zr nsgrg gur gbtc cnecl
4	ldds ld zesdq sgd snrz ozqsx	17	yqqf yq mrfqd ftq fasm bmdfk
5	kccr kc ydrpc rfc rmey nyprw	18	xppe xp lqepc espezrl alcej
6	jbbq jb xcqbo qeb qldx mxoqv	19	wood wo kpdob dro dyqk zkbdi
7	iaap ia wbpap pda pkcw lwnpu	20	vnnc vn jocna cqn expj yjach
8	hzzo hz vaozm ocz ojbv kvmot	21	ummb um inbmz bpm bwoi xizbg
9	gyyn gy uznyl nby niau julns	22	tlla tl hmaly aol avnh whyaf
10	fxxm fx tymxk max mhzt itkmr	23	skkz sk glzlx znk zumg vgxze
11	ewwl ew sxlwj lzw lgys hsjlq	24	rjjy rj fkyjw ymj ytlf ufwyd
12	dvvk dv rwkvi kyv kfxr grikp	25	qiix qi ejxiv xli xske tevxc
13	cuuj cu qvjuh jxu jewq fqhjo		

- Three important characteristics of this problem enabled us to use a brute-force cryptanalysis:
 1. The encryption and decryption algorithms are known.
 2. There are only 25 keys to try.
 3. The language of the plaintext is known and easily recognizable.

Monoalphabetic Ciphers

- Rather than just shifting the alphabet, could shuffle the letters arbitrarily
→ each plaintext letter maps to a different random ciphertext letter.

- The key is 26 letters long:

Plain: abcdefghijklmnopqrstuvwxyz

Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN

Key

Plaintext: ifwewishtoreplaceletters

Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA

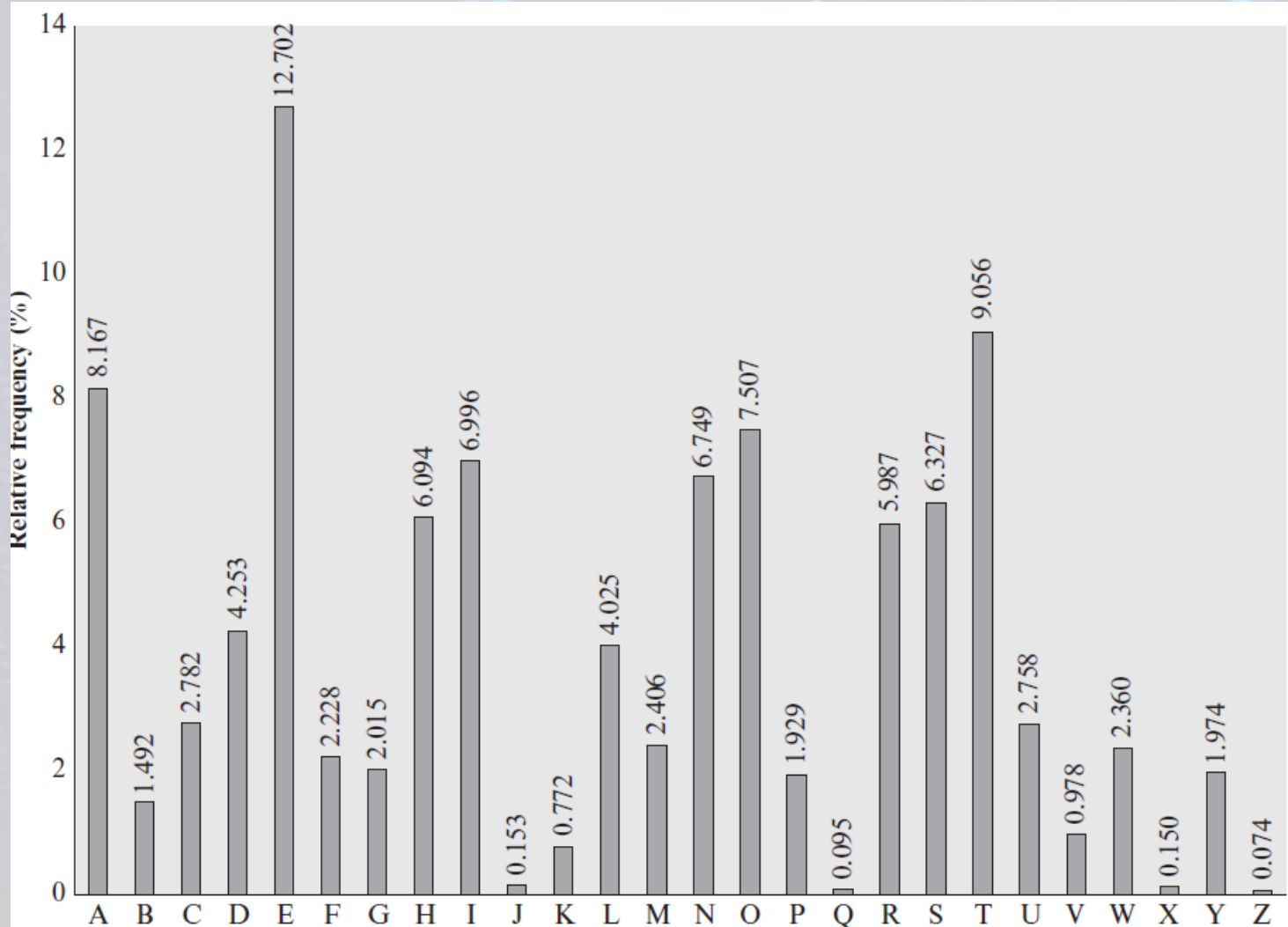
- A dramatic increase in the key space can be achieved by allowing an arbitrary substitution: we have a total of **26!** = 4×10^{26} possible keys

Cryptanalysis of Monoalphabetic Ciphers

- The key space seems to eliminate brute-force techniques for cryptanalysis.
- There is, however, another line of attack: using **language characteristics**.
- If the cryptanalyst knows the nature of the plaintext (e.g., non-compressed English text), then the analyst can exploit the regularities of the language.
- The relative frequency of the letters in the ciphertext can be determined and compared to a **standard frequency distribution** for English: if the message was long enough, this technique alone might be sufficient, but if the message is relatively short, we cannot expect an exact match.

Cryptanalysis of Monoalphabetic Ciphers

- Relative frequency of letters in English text:



- E is by far the most common letter
- followed by T,R,N,I,O,A,S
- other letters like Z,J,K,Q,X are fairly rare

Cryptanalysis of Monoalphabetic Ciphers

- Example: Consider this cipher text: (it is a short message)

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPDTSVPQUZWMYXUZHUSX
EPYEPOPDZSZUFPOMBZWPFPUPZHMDJUDTMOHMQ

- Letters frequency:

P 13.33	H 5.83	F 3.33	B 1.67	C 0.00
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.00
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.00
U 8.33	V 4.17	T 2.50	I 0.83	N 0.00
O 7.50	X 4.17	A 1.67	J 0.83	R 0.00
M 6.67				

- It seems likely that cipher letters *P* and *Z* are the equivalents of plain letters *e* and *t*, but it is not certain which is which.
- The letters *S*, *U*, *O*, *M*, and *H* are all of relatively high frequency and probably correspond to plain letters from the set {a, h, i, n, o, r, s}.
- The letters with the lowest frequencies (namely, *A*, *B*, *G*, *Y*, *I*, *J*) are likely included in the set {b, j, k, q, v, x, z}.

Cryptanalysis of Monoalphabetic Ciphers

- We could make some tentative assignments (trial and error) and start to fill in the plaintext to see if it looks like a reasonable “skeleton” of a message.
- A powerful tool is to look at the frequency of two-letter combinations, known as **digrams**.
- The most common digram in the example is ZW, which appears three times → so we make the correspondence of Z with t and W with h.
- Then, by our earlier hypothesis, we can equate P with e.
- Now notice that the sequence ZWP appears in the ciphertext, and we can translate that sequence as “the.” This is the **most frequent trigram** (three-letter combination) in English, which seems to indicate that we are on the right track.
- Next, notice the sequence ZWSZ in the first line. We do not know that these four letters form a complete word, but if they do, it is of the form th_t. If so, S equates with a.
- So far, we reached:

```
UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
t a           e e te a that e e a           a
VUEPHZHMZSHZOWSFPAPDTSVPQUZWYMXUZHUSX
e t   ta t ha e ee a e th   t a
EPYEPOPDZSZUFPOMBZWPFPUPZHMDJUDTMOHMQ
e e e tat e   the t
```

- Continued analysis of frequencies plus trial and error should easily yield a solution from this point.

Playfair Cipher

- The best-known multiple-letter encryption cipher is the Playfair, which treats *digrams* in the plaintext as single units and translates these units into ciphertext *digrams*.
- Construct a 5x5 matrix: the matrix is filled from *left to right* from *top to bottom*
 - ❑ fill in letters of a keyword (without duplicates)
 - ❑ fill rest of matrix with other letters (without duplicates)
 - ❑ letters I and J count as one letter
- For keyword “*monarchy*”:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Cipher

- Plaintext is encrypted **two letters at a time**, according to the following rules:

- Repeating plaintext letters that are in the same pair are separated with a filler letter, such as x, so that balloon would be treated as ba lx lo on.
- Two plaintext letters that fall in the same row of the matrix are each replaced by the letter to the right, with the first element of the row circularly following the last. For example, ar is encrypted as RM.
- Two plaintext letters that fall in the same column are each replaced by the letter beneath, with the top element of the column circularly following the last. For example, mu is encrypted as CM.
- Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter. Thus, hs becomes BP and ea becomes IM (or JM, as the encipherer wishes).

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Polyalphabetic Ciphers

- Another way to improve on the simple monoalphabetic technique is using different monoalphabetic substitutions as one proceeds through the plaintext message.
- All these techniques have the following features in common:
 1. A set of related monoalphabetic substitution rules is used.
 2. A key determines which particular rule is chosen for a given transformation.

Vigenère Cipher

- A sequence of plaintext letters: $P = p_0, p_1, p_2, \dots, p_{n-1}$
- A key consisting of the sequence of letters: $K = k_0, k_1, k_2, \dots, k_{m-1}$
 - where typically $m < n$.
- The ciphertext letters are calculated as follows:

$$\begin{aligned} C &= C_0, C_1, C_2, \dots, C_{n-1} = E(K, P) = E[(k_0, k_1, k_2, \dots, k_{m-1}), (p_0, p_1, p_2, \dots, p_{n-1})] \\ &= (p_0 + k_0) \bmod 26, (p_1 + k_1) \bmod 26, \dots, (p_{m-1} + k_{m-1}) \bmod 26, \\ &\quad (p_m + k_0) \bmod 26, (p_{m+1} + k_1) \bmod 26, \dots, (p_{2m-1} + k_{m-1}) \bmod 26, \dots \end{aligned}$$

❑ Effectively multiple Caesar ciphers

❑ For the next m letters of the plaintext, the **key letters are repeated**.

- Example:

key:	<i>deceptivedeceptivedeceptive</i>
plaintext:	<i>wearediscoveredsaveyourself</i>
ciphertext:	<i>ZICVTWQNGRZGVTWAVZHCQYGLMGJ</i>

Autokey

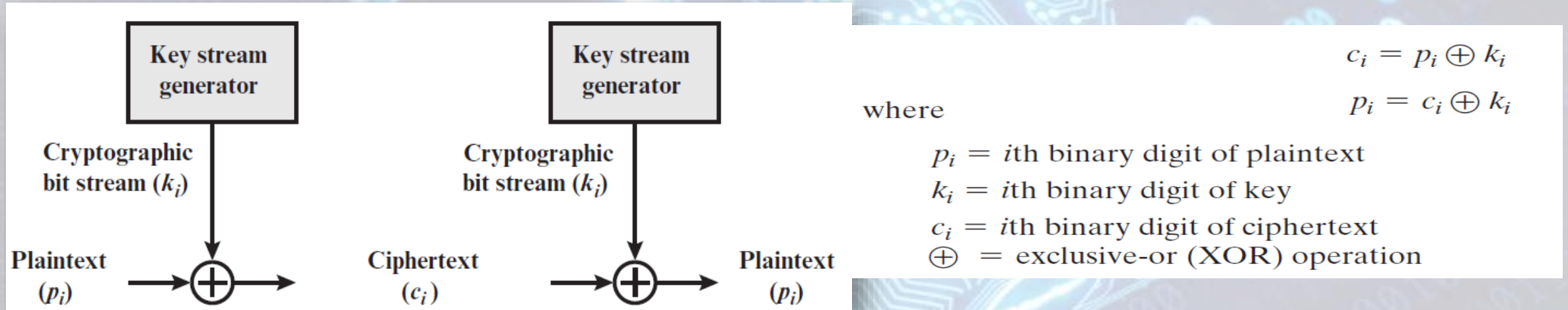
- Similar to Vigenère in which a keyword is **concatenated with the plaintext** itself to provide a running key.

- Example:

key:	<i>deceptivewarediscoveredsav</i>
plaintext:	<i>warediscoveredsaveyourself</i>
ciphertext:	<i>ZICVTWQNGKZEIIGASXSTSLVVWLA</i>

Vernam Cipher

- The ultimate defense against language cryptanalysis is to choose a keyword that is as long as the plaintext and has no statistical relationship to it.
- Works on binary data (bits) rather than letters.



- The essence of this technique is the means of construction of the key.
- Vernam proposed the use of a running loop that eventually repeated the key, so that in fact the system worked with a very long but repeating keyword.

One-Time Pad

- Use a random key that is **as long as the message**, so that the key need not be repeated.
- The key is to be used to encrypt and decrypt a single message, and then is **discarded**.
 - Each new message requires a new key of the same length as the new message.
- Unbreakable: it produces random output that bears no statistical relationship to the plaintext.
- Two different decryptions using two different keys:

```
ciphertext: ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIUERFPLUYTS
key:        pxlmvmsydozufyrvzwc tnlebnecvgdupahfzzlmnyih
plaintext:  mr mustard with the candlestick in the hall

ciphertext: ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIUERFPLUYTS
key:        pftgpmiydgaxgoufhklllmhsqdqogtewbqfgyovuhwt
plaintext:  miss scarlet with the knife in the library
```

space is the 27th letter so
take mod 27 not 26

How to compute mod for
-ve number??

One-Time Pad

- Suppose that a cryptanalyst had managed to find these two keys.
- Two reasonable plaintexts are produced. How is the cryptanalyst to decide which is the correct decryption (i.e., which is the correct key)?
- If the actual keys were produced in a truly random fashion, then the cryptanalyst cannot say that one of these two keys is more likely than the other.
→ Thus, there is no way to decide which key is correct and therefore which plaintext is correct.

Therefore, if you did an **exhaustive search** of all possible keys, you would end up with many legible plaintexts, with no way of knowing which was the intended plaintext.

Therefore, the code is unbreakable.

- The one-time pad offers complete security but, in practice, has two fundamental difficulties:
 - Making large quantities of random keys.
 - The problem of key distribution and protection.

Rail Fence

- The plaintext is written down as a sequence of **diagonals** and then read off as a sequence of **rows**.
- The rail fence depth is the number of rows.
- Example: “meet me after the toga party” with a rail fence of depth 2, we write the following:

m e m a t r h t g p r y
e t e f e t e o a a t

Ciphertext

MEMATRHTGPRYETEFETEOAAT

- Example: cipher → “TEKOOHRACIRMNREATANFTETYTGHH” using key/depth=4.

T				E				K				O				O			
	H			R	A			C	I			R	M			N	R		
		E	A			T	A			N	F			T	E			T	
			Y				T				G				H				H

- Number of rows=depth=4
- Number of columns=ciphertext length=28

Plaintext="they are attacking from the north"

Row Transposition

- A more complex scheme is to write the message in a rectangle “**row by row**” and read the message off “**column by column**”.

→ but permute the order of the columns: this is the key.

- Example:

Key:	4	3	1	2	5	6	7
Plaintext:	a	t	t	a	c	k	p
	o	s	t	p	o	n	e
	d	u	n	t	i	l	t
	w	o	a	m	x	y	z
Ciphertext:	T	T	N	A	P	T	M
	T	S	U	O	A	O	D
	W	C	O	I	X	K	N
	L	P	E	T	Z		

complete missing data
by last letters in
alphabet

Row Transposition

- The transposition cipher can be made significantly more secure by performing **more than one stage** of transposition.
- Example: **1]**

Key:	4 3 1 2 5 6 7
Plaintext:	a t t a c k p o s t p o n e d u n t i l t w o a m x y z
Ciphertext:	TTNAAPTMTSUOAODWCOIXKNLYPETZ

2]

Key:	4 3 1 2 5 6 7
Input:	t t n a a p t m t s u o a o d w c o i x k n l y p e t z
Output:	NSCYAUOPTTWLTMDNAOIEPAXTTOKZ



Thank You